# F70LA Life Insurance Mathematics A (2025/26)

## Tutorial 5

1. *Adverse selection* is the risk that very high-risk individuals will buy insurance in disproportionate numbers to the numbers expected by the insurer.

   Explain why an insurer might demand more rigorous evidence of a prospective policyholder's health status for a term insurance than for a whole life insurance.

2. Lenders offering mortgages to home owners may require the borrower to purchase life insurance to cover the outstanding loan on the death of the borrower, even though the mortgaged property is the loan collateral.

   (a) Explain why the lender might require term insurance in this circumstance.

   (b) Describe how this term insurance might differ from the standard term insurance.

   *Uninsurable lives* are people with such a high risk that the insurer will not sell them an insurance contract at any price. About 2%-3% of applicants for insurance are in this category.

   (c) Can you see any problems with lenders requiring term insurance from borrowers?

3. Some insurers design whole life contracts with premiums payable only up to a high age, such as age 80 years. Why?

4. You are given the following table of values for $\ell_x$ and $A_x$ as the age $x$ increases, assuming an effective rate of interest of 4% per annum.

| $x$ | $\ell_x$ | $A_x$ |
|---|---|---|
| 55 | 99 176.23 | 0.29962 |
| 56 | 99 072.49 | 0.31088 |
| 57 | 98 953.70 | 0.32250 |
| 58 | 98 817.64 | 0.33450 |
| 59 | 98 662.00 | 0.34685 |
| 60 | 98 484.31 | 0.35954 |

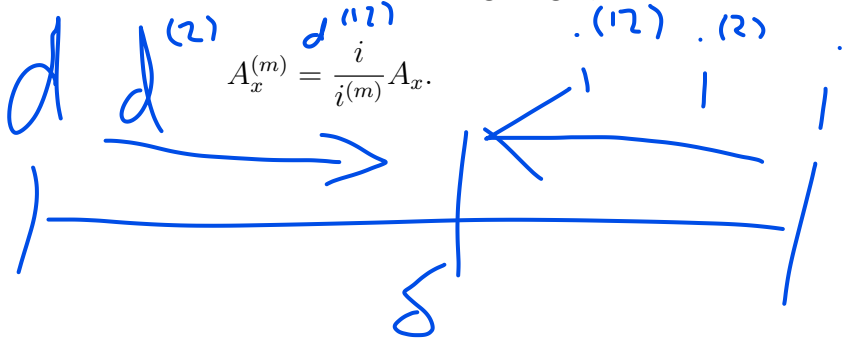
   Calculate the single premiums:

   (a) $100\,000\,{}_5|A_{55}$; (b) $100\,000\,{}_3|A^1_{55:\overline{2}|}$;

   (c) $100\,000\,{}_3|A_{55:\overline{2}|}$.

5. Assuming a uniform distribution of deaths between integer ages, show that for $x \geq 0$ and integer $m > 0$,

$$A_x^{(m)} = \frac{i}{i^{(m)}} A_x.$$



$i^{(m)} = (1+j)$ $A^{(m)} > A$

6. Show that, for $x \geq 0$ and integer $m > 0$, the recursion formula

$$A_x^{(m)} = v^{1/m} \, {}_{\frac{1}{m}}q_x + v^{1/m} \, {}_{\frac{1}{m}}p_x \, A_{x+\frac{1}{m}}^{(m)}$$

holds.

7. Assume that the effective rate of interest is a constant 6% per annum. You are given that

$$A_{40} = 0.25, \qquad A_{60} = 0.4, \qquad A_{40:\overline{20|}} = 0.55.$$

Calculate $100\,000\bar{A}_{40:\overline{20|}}$ using

(a) Claims accceleration.

(b) UDD.

8. You are given the following select life table.

| $x$ | $\ell_{[x]}$ | $\ell_{[x]+1}$ | $\ell_{[x]+2}$ | $\ell_{[x]+3}$ | $\ell_{x+4}$ | $x+4$ |
|---|---|---|---|---|---|---|
| 40 | 100 000 | 99 899 | 99 724 | 99 520 | 99 288 | 44 |
| 41 | 99 802 | 99 689 | 99 502 | 99 283 | 99 033 | 45 |
| 42 | 99 597 | 99 471 | 99 268 | 99 030 | 98 752 | 46 |
| 43 | 99 365 | 99 225 | 99 007 | 98 747 | 98 435 | 47 |
| 44 | 99 120 | 98 964 | 98 726 | 98 429 | 98 067 | 48 |

Consider a four-year term insurance, deferred for one year, issued to a newly selected life age 40, with sum insured £100 000, payable at the end of year of death. Assuming an interest rate of 5% per annum effective, calculate the expected present value of the insurance contract.

9. Let $Z_1$ denote the present value of an $n$-year term insurance that is payable at the end of year of death, issued to $(x)$. Let $Z_2$ denote the present value of a whole life insurance that is also payable at the end of year of death and issued to the same life $(x)$. Assume the sum insured is £1 for both insurances.

Express the covariance of $Z_1$ and $Z_2$ in actuarial functions, simplifying your answer as much as possible.

10. Let the random variable $T_x$ represent the future lifetime of $(x)$.

(a) Describe in words (how much benefit and when is it paid - both the timing and the event that must occur for it to be payable, i.e. the contingent event) the insurance benefits with the present value random variables given below.

(i)
$$Z_1 = \begin{cases} 50v^{T_x} & \text{if } T_x \leq 20, \\ 0 & \text{if } T_x > 20. \end{cases}$$

(ii)
$$Z_2 = \begin{cases} 0 & \text{if } T_x \leq 5, \\ 10v^5 + 20v^{T_x} & \text{if } T_x > 5. \end{cases}$$

(b) Write down in integral form the formula for the expected value for: (i) $Z_1$, and (ii) $Z_2$.

(c) Derive an expression in terms of standard actuarial functions for the expected values of: (i) $Z_1$, and (ii) $Z_2$.

(d) Derive an expression in terms of standard actuarial functions for the covariance of $Z_1$ and $Z_2$.

11. (a) Derive the following the recursion equation for an $n$-year increasing term insurance.

$$(IA)^1_{x:\overline{n}|} = vq_x + vp_x \left( (IA)^{\ 1}_{x+1:\overline{n-1}|} + A^{\ 1}_{x+1:\overline{n-1}|} \right).$$

(b) Hence derive a recursion relationship for $(IA)_x$.

(c) At an annual effective rate of 6% per annum, $(IA)_{50} = 4.99675$, $A_{51} = 0.24905$ and $A^1_{50:\overline{1}|} = 0.00558$. Calculate $(IA)_{51}$.

Numerical solutions: Q4(a) £29 345.40; Q4(b) £284.01; Q4(c) £81 903.29. Q7(a) 55 147.82; Q7(b) 55 148.54. Q8 725.62. Q9. ${}^2A^1_{x:\overline{n}|} - A^1_{x:\overline{n}|}\, A_x$. Q11(c) 5.0731.